

Part 1: Topic Detection and Summarization of News Articles

Generative AI - Assignment 1 Darrsheni Sapovadia (27PGAI0063)

The BBC News Archive has 2,225 articles spread over five categories. The brief asks for the first 30, so that is what I work with here. For each article I use LangChain to do three things: decide the topic, write a short summary, and pull out the key entities. All three results get added back onto the dataframe at the end.

I am calling Groq for the model. The assignment mentions `llama-3.1-8b-instant` but Groq has since retired the Llama models, so I used `openai/gpt-oss-120b` instead. It is on the same free tier and is quick enough for classification and extraction work.

Setup

```
In [1]: import json
import os
import re
import time

import pandas as pd
from dotenv import load_dotenv
from langchain_core.output_parsers import StrOutputParser
from langchain_core.prompts import ChatPromptTemplate
from langchain_groq import ChatGroq

load_dotenv("../.env")

MODEL = os.getenv("LLM_MODEL", "openai/gpt-oss-120b")

# temperature 0 because these are extraction tasks, not creative writing -
# I want the same answers if I run the notebook again.
#
# reasoning_effort took me a while to work out. This model thinks before it
# answers, and the thinking counts towards max_tokens. On the longer articles it
# spent the whole budget reasoning and handed back an empty string, which is why
# my entity column kept coming out blank. Turning the effort down fixes it and
# makes the run a good deal faster too.
llm = ChatGroq(model=MODEL, temperature=0, max_tokens=800, reasoning_effort="low")

print("Model:", MODEL)
```

Model: openai/gpt-oss-120b

```
In [2]: def ask(chain, article, limit=2500):
        """Run a chain on one article, waiting it out if Groq rate limits us.

        The free tier allows 8,000 tokens a minute and news articles are long, so
        being rate limited here is normal rather than exceptional. Groq puts the
        wait it wants in the error message, so I pull that out and use it instead of
        guessing. If it still fails after six goes I let it raise, because a blank
        string coming back would just turn into an empty cell that I might not spot.
        """
        for attempt in range(6):
            try:
                return chain.invoke({"article": article[:limit]})
            except Exception as error:
                message = str(error)
                limited = "rate" in message.lower() or "429" in message
                if not limited or attempt == 5:
                    raise
                suggested = re.search(r"try again in ([\d.]+)s", message)
                time.sleep(float(suggested.group(1)) + 2 if suggested else 20)

        def find_json(text):
            """Grab the JSON object out of a reply and ignore anything around it."""
            found = re.search(r"\{.*\}", text, re.S)
            if not found:
                return {}
            try:
                return json.loads(found.group())
            except json.JSONDecodeError:
                return {}
```

Step 1: Load the dataset

The file is tab separated rather than comma separated, which caught me out the first time. I keep the original category column too so I can check the predictions at the end.

```
In [3]: news = pd.read_csv("../data/bbc-news-data.csv", sep="\t")

        print("Whole dataset:", news.shape)
        news.head(3)
```

Whole dataset: (2225, 4)

```
Out[3]:
```

	category	filename	title	content
0	business	001.txt	Ad sales boost Time Warner profit	Quarterly profits at US media giant TimeWarne...
1	business	002.txt	Dollar gains on Greenspan speech	The dollar has hit its highest level against ...
2	business	003.txt	Yukos unit buyer faces loan claim	The owners of embattled Russian oil giant Yuk...

```
In [4]: df = news.head(30).copy().reset_index(drop=True)
df = df.rename(columns={
    "title": "Title",
    "content": "Article_Text",
    "category": "True_Category",
})
df.insert(0, "Article_ID", df.index + 1)
df = df[["Article_ID", "Title", "Article_Text", "True_Category"]]

print("Working set:", df.shape)
df.head()
```

Working set: (30, 4)

Out[4]:

	Article_ID	Title	Article_Text	True_Category
0	1	Ad sales boost Time Warner profit	Quarterly profits at US media giant TimeWarne...	business
1	2	Dollar gains on Greenspan speech	The dollar has hit its highest level against ...	business
2	3	Yukos unit buyer faces loan claim	The owners of embattled Russian oil giant Yuk...	business
3	4	High fuel prices hit BA's profits	British Airways has blamed high fuel prices f...	business
4	5	Pernod takeover talk lifts Domecq	Shares in UK drinks and food firm Allied Dome...	business

Step 2: Topic classification

I give the model the five categories and a few worked examples before the real article. The examples are mainly there to stop it answering in a full sentence.

The two sentences in the system prompt about companies and the economy were not in my first version. I added them after looking at what it was getting wrong. The BBC files anything to do with trade, prices or the economy under business even when it is a government doing it, and without that steer the model kept calling those articles Politics. That is a perfectly reasonable reading, it just is not the one the dataset uses, and spelling the rule out fixed most of those cases.

```
In [5]: CATEGORIES = ["Business", "Entertainment", "Politics", "Sport", "Tech"]

topic_prompt = ChatPromptTemplate.from_messages([
    ("system",
     "You are a news desk editor. Put each article into exactly one category: "
     "Business, Entertainment, Politics, Sport or Tech.\n"
     "Anything about companies, markets, jobs, trade, prices or the wider economy is "
     "
     "Business, even when governments, courts or regulators are involved. "
     "Keep Politics for stories that are mainly about elections, parties, or the
     running "
     "of government itself.\n"
     "Reply with the category name and nothing else."),
    ("human",
     "Analyze the following news article and identify its topic as one of the
     following "
     "categories: Business, Entertainment, Politics, Sport, or Tech.\n\n"
     "Article: Shares in the airline fell 4% after it warned that fuel costs would
     wipe "
     "out full-year profits."),
    ("ai", "Business"),
    ("human",
     "Article: The government will lift fuel subsidies next month, a move economists
     warn "
     "will push inflation higher and squeeze household budgets."),
    ("ai", "Business"),
    ("human",
     "Article: The home secretary faces a backbench rebellion over plans to
     introduce "
     "compulsory identity cards."),
    ("ai", "Politics"),
    ("human",
     "Article: The striker scored twice in the second half to send his side into the
     "
     "cup final."),
    ("ai", "Sport"),
    ("human", "Article: {article}"),
])

topic_chain = topic_prompt | llm | StrOutputParser()
```

```
In [6]: def classify_topic(article):
    reply = ask(topic_chain, article, limit=1500)
    # every so often it adds a stray word or a full stop, so match against the list
    for category in CATEGORIES:
        if category.lower() in reply.lower():
            return category
    return "Other"
```

```
In [7]: sample = df.loc[0]

print("Title:", sample["Title"])
print("Actual category:", sample["True_Category"])
print("Predicted topic:", classify_topic(sample["Article_Text"]))
```

Title: Ad sales boost Time Warner profit
Actual category: business

Predicted topic: Business

Step 3: Summarization

Two or three sentences, and I tell it to stay with what the article actually says so it does not start editorialising.

```
In [8]: summary_prompt = ChatPromptTemplate.from_messages([
    ("system",
     "You summarise news articles. Cover who, what, when and where. "
     "Stick to what the article says and do not add any opinion of your own."),
    ("human",
     "Summarize the main points of the following news article in 2-3 sentences.\n\n"
     "Article:\n{article}"),
])

summary_chain = summary_prompt | llm | StrOutputParser()

def summarize(article):
    return ask(summary_chain, article).strip()
```

```
In [9]: print("Title:", sample["Title"])
print()
print("Summary:", summarize(sample["Article_Text"]))
```

Title: Ad sales boost Time Warner profit

Summary: TimeWarner's quarterly profit jumped 76% to \$1.13 billion for the three months to December, helped by higher sales of high-speed internet, stronger advertising revenue and one-off gains that offset a dip at Warner Bros and a loss of AOL subscribers. The company's fourth-quarter sales rose 2% to \$11.1 billion, it now owns 8% of Google, and it is preparing to restate its 2000 and 2003 results after an SEC probe into AOL while projecting about 5% operating-earnings growth for 2005.

Step 4: Key entity extraction

I ask for JSON with three groups so the output is easy to parse, then flatten it into one list per article. Asking for plain text here gave me bullet points and headings that were a nuisance to clean up.

```
In [10]: entity_prompt = ChatPromptTemplate.from_messages([
    ("system", "You pull named entities out of news articles. Reply with JSON and nothing else."),
    ("human",
     "From the article below, list the names of any important people, organisations or "
     "places mentioned.\n"
     'Reply in this exact shape: {"people": [], "organisations": [], "places": []}\n'
     "Leave a list empty if there is nothing for it.\n\n"
     "Article:\n{article}"),
])

entity_chain = entity_prompt | llm | StrOutputParser()

def get_entities(article):
    data = find_json(ask(entity_chain, article))
    names = []
    for group in ("people", "organisations", "places"):
        for name in data.get(group, []):
            name = str(name).strip()
            if name and name not in names:
                names.append(name)
    return names

In [11]: print("Title:", sample["Title"])
print()
print("Key entities:", get_entities(sample["Article_Text"]))
```

Title: Ad sales boost Time Warner profit

Key entities: ['Richard Parsons', 'TimeWarner', 'Google', 'AOL', 'Warner Bros', 'US Securities Exchange Commission (SEC)', 'Bertelsmann', 'United States']

Step 5: Run all three tasks over the 30 articles

Three calls per article, so ninety in total. The short sleep keeps us under the tokens-per-minute limit on the free Groq tier.

```
In [12]: topics = []
         summaries = []
         entities = []

         start = time.time()

         for position, row in df.iterrows():
             article = row["Article_Text"]

             topics.append(classify_topic(article))
             summaries.append(summarize(article))
             entities.append(get_entities(article))

             print(f"article {position + 1} of {len(df)} done")
             time.sleep(2)

         print(f"\nAll finished in {time.time() - start:.0f} seconds")
```

article 1 of 30 done

article 2 of 30 done

article 3 of 30 done

article 4 of 30 done

article 5 of 30 done

article 6 of 30 done

article 7 of 30 done

article 8 of 30 done

article 9 of 30 done

article 10 of 30 done

article 11 of 30 done

article 12 of 30 done

article 13 of 30 done

article 14 of 30 done

article 15 of 30 done

article 16 of 30 done

article 17 of 30 done

article 18 of 30 done

article 19 of 30 done

article 20 of 30 done

article 21 of 30 done

article 22 of 30 done

article 23 of 30 done

article 24 of 30 done

article 25 of 30 done

article 26 of 30 done

article 27 of 30 done

article 28 of 30 done

article 29 of 30 done

article 30 of 30 done

All finished in 385 seconds

```
In [13]: df["Detected_Topic"] = topics
df["Summary"] = summaries
df["Key_Entities"] = entities

# the three new columns on their own first
df[["Article_ID", "Detected_Topic", "Summary", "Key_Entities"]].head(10)
```

Out[13]:

	Article_ID	Detected_Topic	Summary	Key_Entities
0	1	Business	TimeWarner's quarterly profit jumped 76% to \$1...	[Richard Parsons, TimeWarner, Google, AOL, War...
1	2	Business	The dollar rose to \$1.2871 per euro, its highe...	[Alan Greenspan, Robert Sinche, Federal Reserv...
2	3	Business	The owners of Russia's former oil giant Yukos,...	[Jamie Firestone, Tim Osborne, Mikhail Khodork...
3	4	Business	British Airways reported a 40% drop in pre-ta...	[Rod Eddington, Mike Powell, Martin Broughton,...
4	5	Business	Allied Domecq's London shares rose about 4% af...	[Allied Domecq, Pernod Ricard, Wall Street Jou...
5	6	Business	Japan's revised data show that the economy gre...	[Heizo Takenaka, Paul Sheard, Lehman Brothers,...
6	7	Business	U.S. firms added 146,000 non-farm jobs in Janu...	[President Bush, Herbert Hoover, Rick Egelton,...
7	8	Business	India's finance minister Palaniappan Chidambar...	[Palaniappan Chidambaram, Gordon Brown, G7, Un...
8	9	Business	Ethiopia's crop output rose to 14.27 million t...	[Henri Josserand, Food and Agriculture Organis...
9	10	Business	A U.S. appeals court rejected the federal gove...	[Clinton, US government, Altria Group, RJ Reyn...

The full dataframe, original columns and new ones together

```
In [14]: pd.set_option("display.max_colwidth", 60)
df
```

Out[14]:

	Article_ID	Title	Article_Text	True_Category	Detected_Topic	Summa
0	1	Ad sales boost Time Warner profit	Quarterly profits at US media giant TimeWarner jumped 7...	business	Business	TimeWarner's quarter profit jumped 76% \$1.13 billion
1	2	Dollar gains on Greenspan speech	The dollar has hit its highest level against the euro i...	business	Business	The dollar rose to \$1.28 per euro, its highest level
2	3	Yukos unit buyer faces loan claim	The owners of embattled Russian oil giant Yukos are to ...	business	Business	The owners of Russian former oil giant Yukos through N
3	4	High fuel prices hit BA's profits	British Airways has blamed high fuel prices for a 40% d...	business	Business	British Airways reported 40% drop in pre-tax profit
4	5	Pernod takeover talk lifts Domecq	Shares in UK drinks and food firm Allied Domecq have ri...	business	Business	Allied Domecq's London shares rose about 4% after Wall
5	6	Japan narrowly escapes recession	Japan's economy teetered on the brink of a technical re...	business	Business	Japan's revised data showed that the economy grew only 0.1%
6	7	Jobs growth still slow in the US	The US created fewer jobs than expected in January, but...	business	Business	U.S. firms added 146,000 non-farm jobs in January, falling
7	8	India calls for fair trade rules	India, which attends the G7 meeting of seven leading in...	business	Business	India's finance minister Palaniappan Chidambaram attend
8	9	Ethiopia's crop	Ethiopia produced 14.27	business	Business	Ethiopia's crop output rose to 14.27 million tonnes in

Article_ID	Title	Article_Text	True_Category	Detected_Topic	Summa
	production up 24%	million tonnes of crops in 2004...			
9	10	Court rejects \$280bn tobacco case	business	Business	A U.S. appeals court rejected the federal government's
10	11	Ask Jeeves tips online ad revival	business	Tech	Ask Jeeves reported near-tripling fourth-quarter 2004
11	12	Indonesians face fuel price rise	business	Business	Indonesia's government weighing a fuel-price increase
12	13	Peugeot deal boosts Mitsubishi	business	Business	Mitsubishi Motors, which has been posting steady sales decline
13	14	Telegraph newspapers axe 90 jobs	business	Business	The Daily and Sunday Telegraph newspapers are axing 90 editorial jobs
14	15	Air passengers win new EU rights	business	Business	EU regulators have introduced new rules that require airlines to
15	16	China keeps tight rein on credit	business	Business	China will keep tightening credit controls into 2005 to
16	17	Parmalat boasts	business	Business	Parmalat, the Italian food group that filed for

Article_ID	Title	Article_Text	True_Category	Detected_Topic	Summa
	doubled profits	food group at the centre of one o...			bankrup
17	18	India's rupee hits five-year high	business	Business	India's rupee rose to five-year high of 43.3 per U.
18	19	India widens access to telecoms	business	Business	India has increased t foreign-direct investme ceiling
19	20	Call centre users 'lose patience'	business	Business	A Dimension Data surv finds that callers a losing p
20	21	Rank 'set to sell off film unit'	business	Business	Leisure group Rank expected to announce demerger o
21	22	Sluggish economy hits German jobs	business	Business	Germany's unemployment rose for the tenth mor in Noven
22	23	Mixed signals from French economy	business	Business	France's econor accelerated in the la quarter of 200.
23	24	US trade gap hits record in 2004	business	Business	The U.S. trade defi reached a recc \$671.7 billion
24	25	Yukos loses US bankruptcy battle	business	Business	A Texas judge dismiss Yukos's bid for Chapter bank

Article_ID	Title	Article_Text	True_Category	Detected_Topic	Summa	
		by Russian oil giant Y...				
25	26	Safety alert as GM recalls cars	The world's biggest carmaker General Motors (GM) is rec...	business	Business	General Motors recalling about 200,0 U.S. vehicles
26	27	Steel firm 'to cut' 45,000 jobs	Mittal Steel, one of the world's largest steel producer...	business	Business	Mittal Steel, which v complete its \$4.5 billi purch
27	28	Strong demand triggers oil rally	Crude oil prices surged back above the \$47 a barrel mar...	business	Business	Crude oil prices ro above \$47 a barrel Thursday af
28	29	UK firm faces Venezuelan land row	Venezuelan authorities have said they will seize land O...	business	Business	Venezuelan offici announced that they v confiscat
29	30	Soaring oil 'hits world economy'	The soaring cost of oil has hit global economic growth,...	business	Business	The OECD's late bi-annual report says th soaring o

One row as JSON

Same shape as the example given in the assignment brief.

```
In [15]: row = df.loc[0]
```

```
print(json.dumps({
    "Article_ID": int(row["Article_ID"]),
    "Title": row["Title"],
    "Article_Text": row["Article_Text"][:200] + "... [excerpt]",
    "Detected_Topic": row["Detected_Topic"],
    "Summary": row["Summary"],
    "Key_Entities": row["Key_Entities"],
}, indent=2))
```

```
{
  "Article_ID": 1,
  "Title": "Ad sales boost Time Warner profit",
  "Article_Text": " Quarterly profits at US media giant TimeWarner jumped 76% to $1.13bn (\u00a3600m) for the three months to December, from $639m year-earlier. The firm, which is now one of the biggest investors in Google,... [excerpt]",
  "Detected_Topic": "Business",
  "Summary": "TimeWarner\u2019s quarterly profit jumped 76% to $1.13\u202fbillion for the three months to December, helped by higher sales of high\u2011speed internet, stronger advertising revenue and one\u2011off gains that offset a dip at Warner\u202fBros and a loss of AOL subscribers. The company\u2019s fourth\u2011quarter sales rose 2% to $11.1\u202fbillion, it now owns 8% of Google, and it is preparing to restate its 2000 and 2003 results after an SEC probe into AOL while projecting about 5% operating\u2011earnings growth for 2005.",
  "Key_Entities": [
    "Richard Parsons",
    "TimeWarner",
    "Google",
    "AOL",
    "Warner Bros",
    "US Securities Exchange Commission (SEC)",
    "Bertelsmann",
    "United States"
  ]
}
```

Sanity check on the topics

The dataset comes with its own category labels, so I can compare my predictions against them. Worth flagging one thing though: the file is sorted by category, so the first 30 rows are all business articles. That makes this a soft test. It tells me whether the classifier is wrongly firing on the other four categories, but it says nothing about whether it can actually pick those four out.

```
In [16]: check = df[["Article_ID", "Title", "True_Category", "Detected_Topic"]].copy()
check["Match"] = check["True_Category"].str.lower() ==
check["Detected_Topic"].str.lower()

print(f"Correct: {check['Match'].sum()} out of {len(check)}")
print()

wrong = check[~check["Match"]]
if len(wrong):
    print("Ones it got wrong:")
    print(wrong[["Title", "True_Category",
"Detected_Topic"]].to_string(index=False))
else:
    print("Every article matched.")
```

Correct: 29 out of 30

Ones it got wrong:

	Title	True_Category	Detected_Topic
Ask Jeeves tips online ad revival		business	Tech

So I took a second sample from further down the file, two articles from each of the five categories, and ran the classifier over those as well. That is a fairer test of whether it can tell the categories apart. The random seed is fixed so this picks the same ten every time.

```
In [17]: mixed = (news.groupby("category", group_keys=False)
               .sample(2, random_state=42)
               .reset_index(drop=True))

results = []
for _, item in mixed.iterrows():
    results.append({
        "Title": item["title"][:42],
        "Actual": item["category"].title(),
        "Predicted": classify_topic(item["content"]),
    })
    time.sleep(2)

spot = pd.DataFrame(results)
spot["Match"] = spot["Actual"].str.lower() == spot["Predicted"].str.lower()

print(f"Correct: {spot['Match'].sum()} out of {len(spot)}")
print()
print(spot.to_string(index=False))
```

Correct: 10 out of 10

	Title	Actual	Predicted	Match
	Christmas sales worst since 1981	Business	Business	True
	US retail sales surge in December	Business	Business	True
	Fry set for role in Hitchhiker's	Entertainment	Entertainment	True
	New York rockers top talent poll	Entertainment	Entertainment	True
	Clarke faces ID cards rebellion	Politics	Politics	True
	Ministers deny care sums 'wrong'	Politics	Politics	True
	Collins named UK Athletics chief	Sport	Sport	True
	IAAF to rule on Greek sprint pair	Sport	Sport	True
	Software watching while you work	Tech	Tech	True
	Net fingerprints combat attacks	Tech	Tech	True

Save the results

```
In [18]: out = df.copy()
out["Key_Entities"] = out["Key_Entities"].apply(lambda names: ", ".join(names))
out.to_csv("../outputs/part1_news_results.csv", index=False)

print("Saved", len(out), "rows to outputs/part1_news_results.csv")
```

Saved 30 rows to outputs/part1_news_results.csv